Ibrahim Khalil Shehada

Computer Engineer · University Lecturer · Machine Learning Researcher (Medical Computer Vision)

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PROFILE

Computer engineer and university lecturer with more than five years and ten semesters of teaching machine learning, data science and software engineering to over 5,000 students. Developer and technical lead of DiaFootCare, an AI wound-monitoring app that is the intervention in a registered clinical trial (NCT07815275). First author of a peer-reviewed comparison of deep-learning optimizers. Completing an M.Sc. in Information Technology and seeking doctoral research in trustworthy medical computer vision.

EDUCATION

M.Sc. Information Technology

2022 – present

Islamic University of Gaza

- All 30 credit hours of coursework completed; thesis plan approved (October 2026).
- Thesis: *From Prediction to Intervention: An Explainable AI-Based Early Warning System for Student Dropout Detection*. Supervisor: Prof. Iyad Alshami.

B.Sc. Computer Engineering

Sep 2015 – Aug 2020

Islamic University of Gaza · 76.5% (Good; ≈ 3.0 / 4.0)

- Coursework: Artificial Intelligence, Data Structures & Algorithms, Probability & Statistics, Software Engineering, Operating Systems.

ACADEMIC EXPERIENCE

Lecturer / Teaching Assistant

Feb 2022 – Apr 2026

Al-Aqsa University, Faculty of Information Technology, Gaza

- Designed and taught Machine Learning, Data Science, Pattern Recognition, Mobile and Cross-Platform Programming, UX/UI Design, IT Project Management and Java.
- Developed course materials, supervised applied student projects and mentored students in technical and research work; strong student evaluations.

Programming Instructor / Trainer

Feb 2023 – Apr 2026

Kick-Lance, Mostaqel, Nawa and Bidaya programmes

- Taught Python, data analysis and web development against measurable learning outcomes; supervised capstone projects.

Teaching Assistant

Feb 2023 – Aug 2023

University of Palestine, Gaza

PROFESSIONAL EXPERIENCE

Freelance Flutter & AI Developer

2021 – 2026

Self-employed · six production applications in healthcare, education and payments

- **DiaFootCare** (2026), developer and technical lead. Smartphone app for diabetic foot ulcer monitoring: U-Net/MobileNetV2 segmentation with attention, CLIP-based tissue classification, calibrated wound measurement and healing tracking; fp16 TFLite models run offline. Intervention in clinical trial NCT07815275 (Oman Ministry of Health, 139 participants).
- **Smart Learning & IQ Improvement** (2025): ML-generated personalised questions targeting each student's weak areas.
- **Teacher Learning Platform** (2024): secure lecture-video platform with download protection and attendance tracking.
- **Bitaqati** (2023): digital driver-card payments built with local authorities to address the cash-change shortage.
- **Delivery** (2023) and **E-commerce** (2022) platforms for local businesses.

Flutter Application Developer

Dec 2021 – Feb 2022

Hexa IT Company, Gaza

Front-End Development Intern

Jun 2019 – Aug 2019

Swift Tech, Gaza

PUBLICATIONS

I. K. Shehada, R. R. Atallah, A. Y. Maghari. “Comparative Study of Deep Learning Optimizers: SGD, Momentum SGD, RMSProp, AMSGrad, Adam, Yogi, and Lion.” *International Journal of Innovative Science and Research Technology*, vol. 10, no. 8, Aug. 2025, IJISRT25AUG007. `peer-reviewed`

I. K. Shehada, L. F. Samhan, B. H. Lubbad. “A Multi-Agent LLM System for Autonomous Detection, Diagnosis and Repair of Micro-Frontend Integration Failures.” Zenodo, Sep. 2026. DOI: 10.5281/zenodo.22677057. `preprint`

Three manuscripts in preparation: two in AI for diabetes care and medical imaging, one in machine learning for genomic data.

PROJECTS

Evolutionary Timetable Scheduling System 2022 – 2024

Hybrid (1+1) evolution strategy with shotgun hill-climbing that produces near-optimal, conflict-reduced faculty timetables at Al-Aqsa University.

Volunteer-Work Department Application 2023 – 2024

Led a three-person team building a Flutter front-end and REST API back-end for the university's volunteer operations; delivered on schedule.

SKILLS

Machine learning	Deep learning, computer vision and segmentation, NLP and LLM agents, classical ML, optimisation, explainable AI (SHAP, counterfactuals), evaluation (Dice, IoU, calibration)
Data & frameworks	Python, PyTorch, TensorFlow/Keras, TFLite, scikit-learn, Hugging Face, Pandas, NumPy, SQL, Matplotlib
Engineering	Flutter/Dart, iOS & Android release, Django, FastAPI, Laravel, REST APIs, Firebase, HTML/CSS/JavaScript, Git, Docker, UX/UI (Figma)
Languages	Arabic (native), English (professional working proficiency)

RESEARCH INTERESTS

Computer vision for healthcare; medical image segmentation; uncertainty, calibration and model reliability; deep-learning optimisation; explainable and fair AI; bioinformatics.

VOLUNTEERING

- **Emergency water distribution, Gaza** (Feb – Aug 2024): planned, monitored and reported distribution across three neighbourhoods.
- **Career-guidance speaker** (2022 – 2026): sessions for students on programming, design and freelancing.

REFERENCES

Prof. Iyad Alshami, PhD
Dean, Faculty of Information Technology
Islamic University of Gaza

Dr. Aiman Ahmed AbuSamra
Professor of Computer Engineering
Islamic University of Gaza

Asst. Prof. Dr. Hazem A. Elbaz
Assistant Vice President for Scientific Research
Al-Aqsa University

Hassan M. Dawoud
Head, Intelligent Systems & Computer Engineering Dept.
Al-Aqsa University

Referees' contact details available on request.